

Real-time observation of the diffusion-limited formation of a cation-molecule complex

Corresponding Author: Professor Henrik Stapelfeldt

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Attachments originally included by the reviewers as part of their assessment can be found at the end of this file.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The paper "Real-time observation of the diffusion-limited formation of a cation-molecule complex" by J.K. Christensen et al. addresses bond formation dynamics, using Li^+ and the benzene dimer as a model system. The authors take advantage of the differing solubilities of Li atoms and benzene in helium droplets to design their initial conditions, allowing them to tune the initial separation distance by varying the droplet diameter. This is a clever approach, although it remains limited to reactions involving alkali ions and molecules.

The results combine experimental measurements with RPMD calculations. The dynamics are triggered by a femtosecond laser that ionizes the metal atom, and the distance between the reaction partners is determined using Coulomb explosion imaging following molecular ionization, detected in coincidence with a state-of-the-art detector. Using a dimer is a smart way to observe ion-molecule contact, as the recoil occurs then between $\text{Li}(\text{Bz})^+$ and Bz^+ .

The authors derive a diffusion coefficient from their measurements that differs from the calculated value.

This paper is clearly written, elegantly tackles an interesting problem, and is accessible to a broad audience. It fits well with Nature Communications, and I am happy to recommend its publication after minor revisions:

- In the introduction, the authors could mention the extensive work on the stereodynamics of bimolecular reactions using crossed molecular beams.
- The discrepancy in the diffusion coefficient could also be explained by the fact that Bz_2 may not be centered in the cluster. Instead, it likely roams around, meaning the first observed coincidences may involve the smallest droplets or the closest dimers to Li. The experimental measurements assume that systems containing both Li and Bz_2 are 100% reactive. However, as seen in the videos, some systems may remain for a long time in the "c" step of the dynamics (invisible to the probe). Why not for ever? especially if Li and Bz_2 are initially far apart. This point could be discussed further.

Reviewer #2

(Remarks to the Author)

The review has been attached in the file 'review_nat.pdf'.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors have answered the reviewers properly and changed the manuscript accordingly.

I propose to accept the manuscript for publication.

Reviewer #2

(Remarks to the Author)

The authors carried out significant modifications in line with my comments, having provided satisfactory answers. I recommend its acceptance for publication in Nature Communications.

Open Access This Peer Review File is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made.

In cases where reviewers are anonymous, credit should be given to 'Anonymous Referee' and the source.

The images or other third party material in this Peer Review File are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

To view a copy of this license, visit <https://creativecommons.org/licenses/by/4.0/>

Reply to Reviewer 1

Reviewer:

The paper "Real-time observation of the diffusion-limited formation of a cation-molecule complex" by J.K. Christensen et al. addresses bond formation dynamics, using Li^+ and the benzene dimer as a model system. The authors take advantage of the differing solubilities of Li atoms and benzene in helium droplets to design their initial conditions, allowing them to tune the initial separation distance by varying the droplet diameter. This is a clever approach, although it remains limited to reactions involving alkali ions and molecules.

The results combine experimental measurements with RPMD calculations. The dynamics are triggered by a femtosecond laser that ionizes the metal atom, and the distance between the reaction partners is determined using Coulomb explosion imaging following molecular ionization, detected in coincidence with a state-of-the-art detector. Using a dimer is a smart way to observe ion-molecule contact, as the recoil occurs then between $\text{Li}(\text{Bz})^+$ and Bz^+ .

The authors derive a diffusion coefficient from their measurements that differs from the calculated value. This paper is clearly written, elegantly tackles an interesting problem, and is accessible to a broad audience. It fits well with Nature Communications, and I am happy to recommend its publication after minor revisions:

Authors response

We appreciate the positive comments very much.

Reviewer

In the introduction, the authors could mention the extensive work on the stereodynamics of bimolecular reactions using crossed molecular beams.

Authors response

We fully agree that past work on stereodynamics of bimolecular reactions and scattering using crossed molecular beams should be mentioned. We believe, however, that it would fit better in the Discussion section where future possibilities enabled by the current work are discussed. Therefore, we have rewritten this part significantly and added a 149-word-long paragraph ('In this connection . . . experimental method is time-resolved') to mention and emphasize the importance of crossed molecular beam studies of stereodynamics, and how our work open new opportunities for insight.

Reviewer

The discrepancy in the diffusion coefficient could also be explained by the fact that Bz_2 may not be centered in the cluster. Instead, it likely roams around, meaning the first observed coincidences may involve the smallest droplets or the closest dimers to Li. The experimental measurements assume that systems containing both Li and Bz_2 are 100% reactive. However, as seen in the videos, some systems may remain for a long time in the "c" step of the dynamics (invisible to the probe). Why not for ever? especially if Li and Bz_2 are initially far apart. This point could be discussed further.

Authors response

We agree that the benzene dimer may not initially be located in the center of the droplets. The delocalization of neutral species has been addressed in theoretical work previously (K. K. Lehmann, Ref. 25), where it was found that for larger droplets, the molecule tends to roam more (be more localized). We believe this also plays a role for our experimental results. As seen in Fig. 3f in the manuscript, the larger droplets show a deviation in t_{total} from the linear diffusion regime. We

attribute this discrepancy to the delocalization of the Bz₂ molecule; in the text (p. 7 l. 33-36). We have revised this sentence and added 64 words to fully cover this.

Regarding the possibility of Li⁺ and Bz₂ not reacting; we do not think this happens. If the dimer happens to be significantly away from the droplet center, the ion may miss it on the first passage of the center but the attraction between Li⁺ and Bz₂ will bring them together later and lead to formation of the Li⁺Bz₂ complex. If we assume that a substantial part of the droplet size distributions resulted in no reaction between Li⁺ and Bz₂, we would not expect the cumulative distribution function, CDF_{LN}, to provide a good fit of the experimental signal Co(Li⁺Bz, Bz⁺). The fact that CDF_{LN}, actually gives very good fit to the data points for all droplet sizes, see Fig. 3 b-e strongly supports that non-reactive trajectories do not occur.

Reply to Reviewer 2

Reviewer

General Comment: This article addresses a highly relevant topic by introducing an experimental method to explore, in real time, a bimolecular reaction, complementing it with ring-polymer molecular dynamics simulations, and paving the way for real-time imaging of stereo-dynamics in ion-molecule reactions. It is nicely shown how a Li⁺ ion, initially located at the surface of a helium nanodroplet, diffuses towards a benzene dimer, initially located inside the liquid drop, after reaching the regime of a ballistic collision, leading to the Li⁺–dimer complex formation. The use of the helium nanodroplet allows to control the initial arrangement of the Li⁺ ion and the benzene dimer. This work complements previous research on a harpoon-type electron transfer between heliophobic Cs and Cs₂ species, initially located at the droplet surface, and a heliophilic fullerene molecule, fully embedded in its interior (DOI: 10.1021/acs.jpcclett.7b01910 and DOI:10.3390/molecules26195783), with the results having explained experimental measurements using electron ionization mass spectroscopy (DOI: 10.1063/1.4967405). In spite of the high relevance of the present work, the manuscript needs a major revision in order to make it more complete, reliable, and interesting. I have included a list of suggestions for improvement below.

Authors response

We appreciate the positive comments. Also, we respect the criticism and suggestions for changes. We have addressed all the points below.

Reviewer

1) Motivation. The introduction could be extended to better motivate the work in a broader perspective of helium nanodroplet-mediated bimolecular collisions. Besides the mentioned research above (DOI: 10.1021/acs.jpcclett.7b01910 and DOI:10.3390/molecules26195783), it could be interesting to mention on the bimolecular collision between superfluid helium nanodroplets with embedded metal atoms and ions and different surfaces such as titanium dioxide-based surfaces and graphene sheets (see, e.g., DOI: 10.1063/1.4916955 and DOI: DOI: 10.1039/D3CP01303K).

Authors response

First of all, we thank the reviewer for bringing our attention to the work on helium droplets co-doped with an alkali atom (or dimer) at the surface and a fullerene in the interior. Those studies are highly relevant for the results presented in our work and should obviously be mentioned. Like the work on crossed beam bimolecular reactions, brought up by reviewer 1, we believe that the mentioning of the alkali-fullerene studies fits better in the end of the manuscript, where we discuss novel opportunities opened by the current findings, rather than in the introduction. Therefore, we have added new text (49 words) at the end of the Conclusion to mention the alkali-fullerene studies and that our new method may provide insight into e.g. electron transfer in such systems. The new

text includes the two references suggested by the reviewer, another one to Krupka, Hauser and de Lara-Castells (2017) as well as to the experimental paper by Renzler et al. (2016).

Regarding the work on deposition of metal-clusters on surfaces via beams of doped helium droplets we respectfully disagree with the reviewer. That work – both the experimental and theoretical – is certainly most interesting but in our opinion, it is topic-wise far away from the femtosecond time-resolved studies of reactions between individual ions and molecules explored in our work. The introduction focuses on previous results on time-resolved studies of bimolecular reactions going back to the pioneering work of Ahmed Zewail in the 1980s. This is, we believe, the most pertinent context for the work presented in our manuscript.

Extending the introduction to mentioning work on doped helium droplets colliding with surfaces or other complex formation or spectroscopic characterization of bimolecular reactions in helium droplets would neither provide a more relevant background nor serve as any extra useful information for the readers in terms of understanding our research results. Therefore, we respectfully keep the introduction as it is but, as said, extend the discussion of the future opportunities opened by our work – to mention the topics suggested by the two reviewers.

Reviewer

2) Intermolecular interactions.

As an addition, it would be recommendable to dissect the physical contributions of the different pair interactions using Symmetry Adapted Perturbation Theory, which is implemented in open-access quantum-chemistry (QC) packages. In particular, it would allow to establish a correlation between the different steps of the ion-dimer formation, and particularly, between the regime of diffusion dynamics and the dominance of the dispersion contribution to the interaction.

Authors response

The regime where dispersion forces dominate is actually the one before the lithium atom is ionized, i.e. a regime where the reaction cannot take place because the two partners are separated from each other and barely interact. Once the lithium atom is ionized, it is polarization forces that cause the cation to enter the droplet, become solvated and then diffuse. This also happens without benzene molecules in the center. The interaction with the benzene molecules is also important, as they will of course trigger the formation of the complex at some later stage. We agree with the reviewer that it is interesting to distinguish the different contributions to the interaction energy in this case and when revisiting the Li^+ -benzene interaction potential (see below), we have now quantified the magnitude of the non-electrostatic contribution.

The results obtained by partitioning the interaction energy clearly show that the electrostatic component (point charge-quadrupole) is essential in explaining why the cation- π complex is so stable as the cation approaches the benzene dimer along the reaction coordinate, perpendicular to the dimer plane. Parallel interactions, on the other hand, barely exhibit any contribution of the charge-quadrupole term, as expected.

Reviewer

As presented in the supplementary information, the Li-dimer interaction is taken as a sum of pair potentials of the 6-12 Lennard-Jones (LJ)-type, with the parameters having been fitted to QC calculations using the coupled cluster approach with single, double, and perturbative triples excitation (i.e., the CCSD(T) method). Although the CCSD(T) method is an accurate approach for this type of interaction, the used electronic basis set, the augmented polarized correlation-consistent double- ζ basis set, is not large enough to describe it precisely, specially at long-range. It would be thus convenient that the authors extend their computations with the triple- ζ basis set, including an

extrapolation to the complete basis set limit (see, e.g., DOI: 10.1063/1.1415078) in order to explain how much the basis set enlargement affects the fitting parameters.

Authors response

The interaction between neutral Li and the benzene dimer is extremely weak at the long distances relevant prior to ionization. The interaction at arbitrary ranges is more complex but not of interest in our work because we only consider the Li atom at the droplet surface where it is very far (tens of Å) from the benzene dimer. The Lennard-Jones form that we chose is consistent with the main interaction in the systems, which are also of the LJ form (especially among helium atoms), but should only be considered as a compromise between computational efficiency and chemical accuracy. The CCSD(T) calculations that we performed were limited to the long-range region as well - that is, for distances above 10 Å. The few points selected along the Z and X axes were used to fit the parameters of the LJ potential between Li and C and H atoms of the benzene monomer, in the region well above the equilibrium distance of a few Å.

It is true that the double zeta basis set remains relatively modest, however we could not achieve quadruple zeta quality due to excessive memory requirements. Two calculations at the triple zeta level could be successfully performed, for Li-benzene distances of 10 Å and 20 Å along the Z axis, with the values given in the table below, in comparison with the double zeta basis set.

Z (Å)	aug-cc-pVDZ (meV)	aug-cc-pVTZ (meV)	AVTZ-AVDZ (meV)
10	0.9265	0.7031	-0.2234
20	0.00843	0.00843	<0.00001

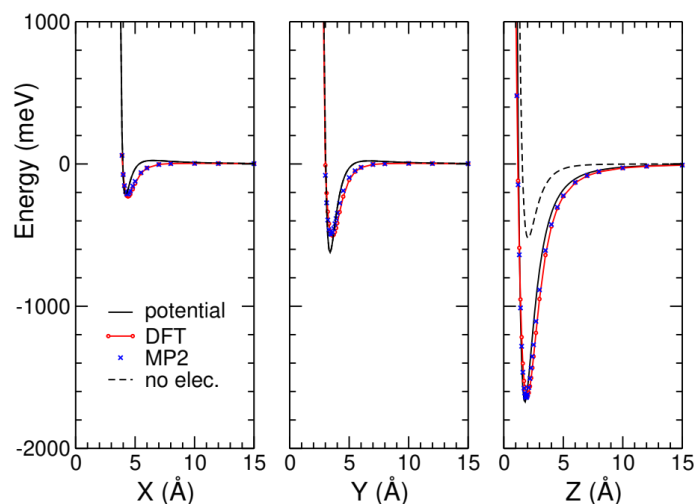
Based on these new calculations, and without attempting to extrapolate to the complete basis set limit, we can see that for Li-benzene distances larger than 20 Å (which is always the case in the experiments and simulations) the difference between the results for the two basis sets is negligible. Thus, we can safely conclude that the Li-benzene interaction remains extremely weak and would not affect the equilibrium properties of the droplet, prior to ionization. Actually, with magnitudes of the interaction of a few micro-eV, we believe that neglecting the Li-benzene interactions altogether in these equilibrium simulations would not alter the subsequent results after ionization. In our revision, we have further mentioned the additional calculations performed for Li-benzene in the Supplementary Information and commented on the relevance of the resulting interaction for droplets containing 3600 helium atoms or more.

Reviewer

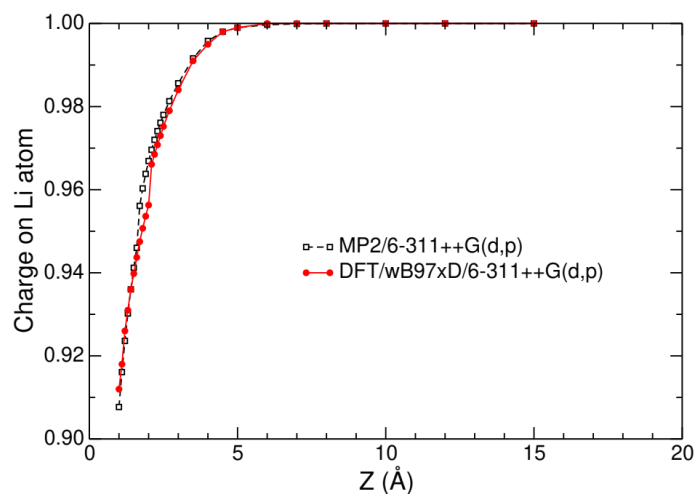
As very well explained in the supplementary information, the interaction between the Li⁺ ion and the benzene dimer, it is much more complex, involving a partial charge transfer upon the cation- π complex formation. The inter-atomic pairwise functional form is correct although it would be advisable to use more accurate potential functions that have been previously applied to complexes formation implying charge transfer and other strong attractive energy contributions beyond van-der-Waals forces (see, e.g., DOI: 10.1039/D4RA05401F). It would be also desirable to apply higher levels of ab initio theory either to fit the potential parameters or to validate the performance of the Density Functional Theory approaches that have been considered. If the application of the CCSD(T) approach is computationally too demanding, the the domain-based pair natural orbital coupled-cluster approach DLPNO-CCSD(T) could be a feasible option (see, e.g., DOI: 10.1063/1.4773581). A practical option could also be the application of the coupled Möller Plesset second-order perturbation theory method (see, e.g., DOI: 10.1063/1.2905808, and DOI: 10.1021/ct9005882).

Authors response

The discussion in the Supplementary Information refers to charge transfer in the dicationic case (a second ionization is performed to trigger Coulomb explosion and record the charged fragments). The purpose of this calculation was to show that in the dicationic complex, one charge localized on the benzene molecule furthest from Li^+ and the second on the Li^+Bz complex. The resulting Coulomb explosion would then yield Li^+Bz and Bz^+ ions, which is what we observe. However, we agree that we did not discuss at length the case of the cationic case (singly-charge), which is perhaps even more important as it is the one where the reaction takes place. For cation- π complexes, DFT turns out to capture the interaction rather well, as discussed in earlier papers of such compounds [Ref. 46-47 of manuscript + ref. 5 of SI], and we did not consider coupled cluster approaches to such a problem. However, to support our statement, we have performed additional quantum chemistry calculations, using MP2 instead of DFT, but keeping the basis set as 6-311++G(d,p). The results, now included in Supplementary Fig. 2, are very close to the DFT data used to parametrize the potential, at least within the natural error associated with the simple functional form used for this potential. The resulting data are also shown below, with the electrostatic contribution removed from the potential energy depicted with dashed lines:



Concerning the specific charge transfer problem, also in relation to the potential itself, there is essentially no charge transfer between Li^+ and the benzene monomer upon complex formation. The figure below gives the variations of the effective charge on the lithium atom along the Z coordinate, as obtained from a natural bond orbital population analysis:



The residual electronic fraction transferred from the Li^+ to the benzene molecule thus amounts to less than 0.05 electron at the equilibrium position, which is well accounted for by polarization forces.

Finally, and while we agree with the reviewer that more sophisticated forms for the Li^+ -benzene interaction models could always be designed (we added a reference to Krupka, Krezminska and de Lara-Castells), they would likely not change the mechanisms observed in our numerical experiments, nor the quantitative reaction times or diffusion velocities – but they would make the simulations quite more expensive.

Reviewer

3) Diffusion dynamics and friction.

In page 7 of the main manuscript (line 25), it can be read ‘Our simulations show that the linear motion results from the combined effect of Li^+ being accelerated by the attractive interaction with the He atoms and Bz2 and decelerated by friction from the droplet’. This is a very engaging observation that can be connected with findings in the previous research on the spatial quenching of a bimolecular collision due the reaction barrier stemming from the extrusion of the helium droplet upon approach of the two reactants (DOI:10.1039/C6CP06858H). As a future prospect, it would be instructive to use the same approach as a complement to the ring-polymer molecular dynamics simulations since it accounts for the superfluid nature of the quantum fluid through the application of helium density functional theory.

Authors response

We agree with the reviewer that the He-(TD)DFT approach would be a natural step beyond the present work. At the moment, it is unclear whether this approach can treat droplets of approximately 4000 atoms, over simulation times of several tens of picosecond and with a minimum account of statistics, but it is definitely an effort worth considering. In the revision, we have added a sentence in the concluding sections to motivate such an effort, together with the appropriate reference.

Reviewer

Minor Comment. Please include the coordinates of the most relevant complexes at the lowest-energy configurations for the sake of reproducibility.

Authors response

We now provide the equilibrium geometries of the Li^+ -benzene and Li^+ -benzene dimer (side and sandwich forms) cation- π complexes, as obtained from our DFT optimizations, in the Supplementary Information.

“Real-time observation of the diffusion-limited formation of a cation-molecule complex”

by Jeppe K. Christensen, Christian Engelbrecht Petersen, Simon. H. Albrechtsen, Jean Goudot,
Florent Calvo, Henrik Stapelfeldt

Recommendation: Major revision (a further review might be required).

General Comment: This article addresses a highly relevant topic by introducing an experimental method to explore, in real time, a bimolecular reaction, complementing it with ring-polymer molecular dynamics simulations, and paving the way for real-time imaging of stereo-dynamics in ion-molecule reactions. It is nicely shown how a Li^+ ion, initially located at the surface of a helium nanodroplet, diffuses towards a benzene dimer, initially located inside the liquid drop, after reaching the regime of a ballistic collision, leading to the Li^+ -dimer complex formation. The use of the helium nanodroplet allows to control the initial arrangement of the Li^+ ion and the benzene dimer. This work complements previous research on a harpoon-type electron transfer between heliophobic Cs and Cs_2 species, initially located at the droplet surface, and a heliophilic fullerene molecule, fully embedded in its interior (DOI: 10.1021/acs.jpcllett.7b01910 and DOI:10.3390/molecules26195783), with the results having explained experimental measurements using electron ionization mass spectroscopy (DOI: 10.1063/1.4967405). In spite of the high relevance of the present work, the manuscript needs a major revision in order to make it more complete, reliable, and interesting. I have included a list of suggestions for improvement below.

1) Motivation. The introduction could be extended to better motivate the work in a broader perspective of helium nanodroplet-mediated bimolecular collisions. Besides the mentioned research above (DOI: 10.1021/acs.jpcllett.7b01910 and DOI:10.3390/molecules26195783), it could be interesting to mention on the bimolecular collision between superfluid helium nanodroplets with embedded metal atoms and ions and different surfaces such as titanium dioxide-based surfaces and graphene sheets (see, e.g., DOI: 10.1063/1.4916955 and DOI: DOI: 10.1039/D3CP01303K).

2) Intermolecular interactions.

- As an addition, it would be recommendable to dissect the physical contributions of the different pair interactions using Symmetry Adapted Perturbation Theory, which is implemented in open-access quantum-chemistry (QC) packages. In particular, it would allow to establish a correlation between the different steps of the ion-dimer formation, and particularly, between the regime of diffusion dynamics and the dominance of the dispersion contribution to the interaction.
- As presented in the supplementary information, the Li-dimer interaction is taken as a sum of pair potentials of the 6-12 Lennard-Jones (LJ)-type, with the parameters having been fitted to QC calculations using the coupled cluster approach with single, double, and perturbative triples excitation (i.e., the CCSD(T) method). Although the CCSD(T) method is an accurate approach for this type of interaction, the used electronic basis set, the augmented polarized correlation-consistent double- ζ basis set, is not large enough to describe it precisely, specially at long-range. It would be thus convenient that the authors extend their computations with the triple- ζ basis set, including an extrapolation to the complete basis set limit (see, e.g., DOI: 10.1063/1.1415078) in order to explain how much the basis set enlargement affects the fitting parameters.

- As very well explained in the supplementary information, the interaction between the Li^+ ion and the benzene dimer, it is much more complex, involving a partial charge transfer upon the cation- π complex formation. The inter-atomic pairwise functional form is correct although it would be advisable to use more accurate potential functions that have been previously applied to complexes formation implying charge transfer and other strong attractive energy contributions beyond van-der-Waals forces (see, e.g., DOI: 10.1039/D4RA05401F). It would be also desirable to apply higher levels of *ab initio* theory either to fit the potential parameters or to validate the performance of the Density Functional Theory approaches that have been considered. If the application of the CCSD(T) approach is computationally too demanding, the the domain-based pair natural orbital coupled-cluster approach DLPNO-CCSD(T) could be a feasible option (see, e.g., DOI: 10.1063/1.4773581). A practical option could also be the application of the coupled Möller Plesset second-order perturbation theory method (see, e.g., DOI: 10.1063/1.2905808, and DOI: 10.1021/ct9005882).

3) Diffusion dynamics and friction. In page 7 of the main manuscript (line 25), it can be read ‘Our simulations show that the linear motion results from the combined effect of Li^+ being accelerated by the attractive interaction with the He atoms and Bz_2 and decelerated by friction from the droplet’. This is a very engaging observation that can be connected with findings in the previous research on the spatial quenching of a bimolecular collision due the reaction barrier stemming from the extrusion of the helium droplet upon approach of the two reactants (DOI: 10.1039/C6CP06858H). As a future prospect, it would be instructive to use the same approach as a complement to the ring-polymer molecular dynamics simulations since it accounts for the superfluid nature of the quantum fluid through the application of helium density functional theory.

Minor Comment. Please include the coordinates of the most relevant complexes at the lowest-energy configurations for the sake of reproducibility.